LEVEL 1: BASIC ARRAYS

1. Find the largest element

```
#include<stdio.h>
int main(){
int n,i,max;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
max=a[0];
for(i=1;i<n;i++){
if(a[i]>max)max=a[i];}
printf("%d",max); }
```

2. Find the smallest element

```
#include<stdio.h>
int main(){
int n,i,min;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
min=a[0];
for(i=1;i<n;i++){
if(a[i]<min)min=a[i];}
printf("%d",min);}
```

3. Find the second largest element

```
#include<stdio.h>
#include<limits.h>
int main(){
int n,i,max=INT_MIN,second=INT_MIN;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
if(a[i]>max){
second=max;
max=a[i];
}else if(a[i]>second&&a[i]!=max){
second=a[i];}}
printf("%d",second);}
```

4. Find the second smallest element

```
#include<stdio.h>
#include<limits.h>
int main(){
int n,i,min=INT_MAX,second=INT_MAX;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
if(a[i]<min){
second=min;
min=a[i];
}else if(a[i]<second&&a[i]!=min){
second=a[i]; }}
printf("%d",second);}
```

5. Find the sum of all elements

```
#include<stdio.h>
int main(){
int n,i,sum=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++){
scanf("%d",&a[i]);
sum+=a[i]; }
printf("%d",sum); }
```

6. Find the average

```
#include<stdio.h>
int main(){
int n,i,sum=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++){
scanf("%d",&a[i]);
sum+=a[i]; }
printf("%.2f",(float)sum/n); }
```

7. Count even and odd numbers

```
#include<stdio.h>
int main(){
int n,i,even=0,odd=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++){
scanf("%d",&a[i]);
if(a[i]%2==0)even++;
else odd++;
}
printf("Even=%d Odd=%d",even,odd);
}
```

8. Count positive, negative and zero

```
#include<stdio.h>
int main(){
int n,i,p=0,ne=0,z=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++){
scanf("%d",&a[i]);
if(a[i]>0)p++;
else if(a[i]<0)ne++;
```

```
else z++;
}
printf("Positive=%d Negative=%d Zero=%d",p,ne,z);
}
```

9. Print array in reverse

```
#include<stdio.h>
int main(){
int n,i;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=n-1;i>=0;i--)printf("%d ",a[i]);
}
```

10. Reverse an array without another array

```
#include<stdio.h>
int main(){
int n,i,temp;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n/2;i++){
temp=a[i];
a[i]=a[n-1-i];
a[n-1-i]=temp;
}
for(i=0;i<n;i++)printf("%d ",a[i]);
}
```

11. Copy one array into another

```
#include<stdio.h>
int main(){
int n,i;
scanf("%d",&n);
int a[n],b[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++)b[i]=a[i];
for(i=0;i<n;i++)printf("%d ",b[i]);
}
```

12. Find length of array

```
#include<stdio.h>
int main(){
int n,i;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
i=0;
while(i<n)i++;
printf("%d",i);
}
```

13. Linear search

```
#include<stdio.h>
int main(){
int n,i,x,pos=-1;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
scanf("%d",&x);
for(i=0;i<n;i++){
if(a[i]==x){
pos=i;
break;
}
}
printf("%d",pos);
}
```

14. Frequency of a given element

```
#include<stdio.h>
int main(){
int n,i,x,count=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
scanf("%d",&x);
for(i=0;i<n;i++){
if(a[i]==x)count++;
}
printf("%d",count);
}
```

15. Frequency of every element

```
#include<stdio.h>
int main(){
int n,i,j,count;
scanf("%d",&n);
int a[n],visited[n];
for(i=0;i<n;i++){
scanf("%d",&a[i]);
visited[i]=0;
}
for(i=0;i<n;i++){
if(visited[i])continue;
count=1;
for(j=i+1;j<n;j++){
if(a[i]==a[j]){
count++;
visited[j]=1;
}
}
printf("%d=%d\n",a[i],count);
}
}
```

16. Count duplicate elements

```
#include<stdio.h>
int main(){
```

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```
int n,i,j,count,duplicates=0;
scanf("%d",&n);
int a[n],visited[n];
for(i=0;i<n;i++){
scanf("%d",&a[i]);
visited[i]=0;
}
for(i=0;i<n;i++){
if(visited[i])continue;
count=0;
for(j=i+1;j<n;j++){
if(a[i]==a[j]){
count++;
visited[j]=1;
}
}
if(count>0)duplicates++;
}
printf("%d",duplicates);
}
```

17. Print duplicate elements

```
#include<stdio.h>
int main(){
int n,i,j,count;
scanf("%d",&n);
int a[n],visited[n];
for(i=0;i<n;i++){
scanf("%d",&a[i]);
visited[i]=0;
}
for(i=0;i<n;i++){
if(visited[i])continue;
count=0;
for(j=i+1;j<n;j++){
if(a[i]==a[j]){
count++;
visited[j]=1;
}
}
if(count>0)printf("%d ",a[i]);
}
}
```

18. Print unique elements

```
#include<stdio.h>
int main(){
int n,i,j,count;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
count=0;
for(j=0;j<n;j++){
if(a[i]==a[j])count++;
}
if(count==1)printf("%d ",a[i]);
}
}
```

19. Remove duplicate elements

```
#include<stdio.h>
int main(){
int n,i,j,k=0,found;
scanf("%d",&n);
int a[n],b[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
found=0;
for(j=0;j<k;j++){
if(a[i]==b[j])found=1;
}
if(!found)b[k++]=a[i];
}
for(i=0;i<k;i++)printf("%d ",b[i]);
}
```

20. Check ascending order

```
#include<stdio.h>
int main(){
int n,i,sorted=1;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=1;i<n;i++){
if(a[i]<a[i-1]){
sorted=0;
break;
}
}
if(sorted)printf("Sorted");
else printf("Not Sorted");
}
```

LEVEL 2: BASIC LOGIC BUILDING

21. First repeating element

```
#include<stdio.h>
int main(){
int n,i,j,x=-1;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
for(j=i+1;j<n;j++){
if(a[i]==a[j]){
x=a[i];
break;
```

```
}
}
if(x!=-1)break;
}
printf("%d",x);
}
```

22. First non-repeating element

```
#include<stdio.h>
int main(){
int n,i,j,count,x=-1;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
count=0;
for(j=0;j<n;j++){
if(a[i]==a[j])count++;
}
if(count==1){
x=a[i];
break;
}
}
printf("%d",x);
}
```

23. Missing number from 1 to N

```
#include<stdio.h>
int main(){
int n,i,x,sum=0;
scanf("%d",&n);
for(i=0;i<n-1;i++){
scanf("%d",&x);
sum+=x;
}
printf("%d",n*(n+1)/2-sum);
}
```

24. Duplicate number from 1 to N

```
#include<stdio.h>
int main(){
int n,i,j;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
for(j=i+1;j<n;j++){
if(a[i]==a[j]){
printf("%d",a[i]);
return 0;
}
}
}
```

```
}
```

25. Two numbers with target sum

```
#include<stdio.h>
int main(){
int n,i,j,target;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
scanf("%d",&target);
for(i=0;i<n;i++){
for(j=i+1;j<n;j++){
if(a[i]+a[j]==target){
printf("%d %d",a[i],a[j]);
return 0;
}
}
}
}
```

26. Three numbers with target sum

```
#include<stdio.h>
int main(){
int n,i,j,k,target;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
scanf("%d",&target);
for(i=0;i<n;i++){
for(j=i+1;j<n;j++){
for(k=j+1;k<n;k++){
if(a[i]+a[j]+a[k]==target){
printf("%d %d %d",a[i],a[j],a[k]);
return 0;
}
}
}
}
}
```

27. Difference between maximum and minimum

```
#include<stdio.h>
int main(){
int n,i,max,min;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
max=min=a[0];
for(i=1;i<n;i++){
if(a[i]>max)max=a[i];
if(a[i]<min)min=a[i];
}
printf("%d",max-min);
}
```

28. Largest difference a[j]-a[i]

```
#include<stdio.h>
int main(){
int n,i,min,d;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
min=a[0];
d=a[1]-a[0];
for(i=1;i<n;i++){
if(a[i]-min>d)d=a[i]-min;
if(a[i]<min)min=a[i];
}
printf("%d",d);
}
```

29. Smallest difference between any two

```
#include<stdio.h>
#include<stdlib.h>
int main(){
int n,i,j,d=2147483647,x;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
for(j=i+1;j<n;j++){
x=abs(a[i]-a[j]);
if(x<d)d=x;
}
}
printf("%d",d);
}
```

30. Pair with maximum sum

```
#include<stdio.h>
int main(){
int n,i,j,sum=-2147483647,x,y;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
for(j=i+1;j<n;j++){
if(a[i]+a[j]>sum){
sum=a[i]+a[j];
x=a[i];
y=a[j];
}
}
}
printf("%d %d",x,y);
}
```

31. Pair with minimum sum

```
#include<stdio.h>
int main(){
int n,i,j,sum=2147483647,x,y;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
for(j=i+1;j<n;j++){
if(a[i]+a[j]<sum){
sum=a[i]+a[j];
x=a[i];
y=a[j];
}
}
}
printf("%d %d",x,y);
}
```

32. All pairs with target sum

```
#include<stdio.h>
int main(){
int n,i,j,target;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
scanf("%d",&target);
for(i=0;i<n;i++){
for(j=i+1;j<n;j++){
if(a[i]+a[j]==target)printf("%d %d\n",a[i],a[j]);
}
}
}
```

33. All pairs with target difference

```
#include<stdio.h>
#include<stdlib.h>
int main(){
int n,i,j,target;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
scanf("%d",&target);
for(i=0;i<n;i++){
for(j=i+1;j<n;j++){
if(abs(a[i]-a[j])==target)printf("%d %d\n",a[i],a[j]);
}
}
}
```

34. Element occurring only once

```
#include<stdio.h>
int main(){
int n,i,j,count;
scanf("%d",&n);
int a[n];
```

```
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
count=0;
for(j=0;j<n;j++){
if(a[i]==a[j])count++;
}
if(count==1){
printf("%d",a[i]);
return 0;
}
}
}
```

35. Element occurring odd number of times

```
#include<stdio.h>
int main(){
int n,i,j,count;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
count=0;
for(j=0;j<n;j++){
if(a[i]==a[j])count++;
}
if(count%2!=0){
printf("%d",a[i]);
return 0;
}
}
}
```

36. Majority element

```
#include<stdio.h>
int main(){
int n,i,j,count;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
count=0;
for(j=0;j<n;j++){
if(a[i]==a[j])count++;
}
if(count>n/2){
printf("%d",a[i]);
return 0;
}
}
printf("-1");
}
```

37. Elements occurring more than N/2 times

```
#include<stdio.h>
int main(){
int n,i,j,count;
scanf("%d",&n);
int a[n],visited[n];
for(i=0;i<n;i++){
scanf("%d",&a[i]);
visited[i]=0;
}
for(i=0;i<n;i++){
if(visited[i])continue;
count=0;
for(j=i;j<n;j++){
if(a[i]==a[j]){
count++;
visited[j]=1;
}
}
if(count>n/2)printf("%d ",a[i]);
}
}
```

38. Elements occurring more than N/3 times

```
#include<stdio.h>
int main(){
int n,i,j,count;
scanf("%d",&n);
int a[n],visited[n];
for(i=0;i<n;i++){
scanf("%d",&a[i]);
visited[i]=0;
}
for(i=0;i<n;i++){
if(visited[i])continue;
count=0;
for(j=i;j<n;j++){
if(a[i]==a[j]){
count++;
visited[j]=1;
}
}
if(count>n/3)printf("%d ",a[i]);
}
}
```

39. Intersection of two arrays

```
#include<stdio.h>
int main(){
int n,m,i,j,found;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
scanf("%d",&m);
int b[m];
```

```
for(i=0;i<m;i++)scanf("%d",&b[i]);
for(i=0;i<n;i++){
found=0;
for(j=0;j<m;j++){
if(a[i]==b[j]){
found=1;
break;
}
}
if(found)printf("%d ",a[i]);
}
}
```

40. Union of two arrays

```
#include<stdio.h>
int main(){
int n,m,i,j,k=0,found;
scanf("%d",&n);
int a[n+m];
for(i=0;i<n;i++)scanf("%d",&a[i]);
k=n;
scanf("%d",&m);
for(i=0;i<m;i++)scanf("%d",&a[k++]);
for(i=0;i<k;i++){
found=0;
for(j=0;j<i;j++){
if(a[i]==a[j]){
found=1;
break;
}
}
if(!found)printf("%d ",a[i]);
}
}
```

LEVEL 3: ARRAY MANIPULATION

41. Move all zeros to the end

```
#include<stdio.h>
int main(){
int n,i,k=0,temp;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
if(a[i]!=0){
temp=a[i];
a[i]=a[k];
a[k]=temp;
k++;
}
}
for(i=0;i<n;i++)printf("%d ",a[i]);
}
```

42. Move all zeros to the beginning

```
#include<stdio.h>
int main(){
int n,i,k=n-1,temp;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=n-1;i>=0;i--){
if(a[i]!=0){
temp=a[i];
a[i]=a[k];
a[k]=temp;
k--;
}
}
for(i=0;i<n;i++)printf("%d ",a[i]);
}
```

43. Move negative numbers to one side

```
#include<stdio.h>
int main(){
int n,i,k=0,temp;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
if(a[i]<0){
temp=a[i];
a[i]=a[k];
a[k]=temp;
k++;
}
}
for(i=0;i<n;i++)printf("%d ",a[i]);
}
```

44. Separate even and odd numbers

```
#include<stdio.h>
int main(){
int n,i,k=0,temp;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
if(a[i]%2==0){
temp=a[i];
a[i]=a[k];
a[k]=temp;
k++;
}
}
for(i=0;i<n;i++)printf("%d ",a[i]);
}
```

45. Rearrange positive and negative alternately

```
#include<stdio.h>
int main(){
int n,i,j,temp;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
if((i%2==0&&a[i]<0)||(i%2!=0&&a[i]>=0)){
j=i+1;
while(j<n&&((i%2==0&&a[j]<0)||(i%2!=0&&a[j]>=0)))j
++;
if(j<n){
temp=a[i];
a[i]=a[j];
a[j]=temp;
}
}
}
for(i=0;i<n;i++)printf("%d ",a[i]);
}
```

46. Sort array containing only 0, 1 and 2

```
#include<stdio.h>
int main(){
int n,i,zero=0,one=0,two=0,x;
scanf("%d",&n);
for(i=0;i<n;i++){
scanf("%d",&x);
if(x==0)zero++;
else if(x==1)one++;
else two++;
}
for(i=0;i<zero;i++)printf("0 ");
for(i=0;i<one;i++)printf("1 ");
for(i=0;i<two;i++)printf("2 ");
}
```

47. Sort without sorting function

```
#include<stdio.h>
int main(){
int n,i,j,temp;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n-1;i++){
for(j=i+1;j<n;j++){
if(a[i]>a[j]){
temp=a[i];
a[i]=a[j];
a[j]=temp;
}
}
}
for(i=0;i<n;i++)printf("%d ",a[i]);
}
```

48. Rotate left by one position

```
#include<stdio.h>
int main(){
int n,i,temp;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
temp=a[0];
for(i=0;i<n-1;i++)a[i]=a[i+1];
a[n-1]=temp;
for(i=0;i<n;i++)printf("%d ",a[i]);
}
```

49. Rotate right by one position

```
#include<stdio.h>
int main(){
int n,i,temp;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
temp=a[n-1];
for(i=n-1;i>0;i--)a[i]=a[i-1];
a[0]=temp;
for(i=0;i<n;i++)printf("%d ",a[i]);
}
```

50. Rotate left by K positions

```
#include<stdio.h>
int main(){
int n,k,i,j,temp;
scanf("%d%d",&n,&k);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
k%=n;
for(j=0;j<k;j++){
temp=a[0];
for(i=0;i<n-1;i++)a[i]=a[i+1];
a[n-1]=temp;
}
for(i=0;i<n;i++)printf("%d ",a[i]);
}
```

51. Rotate right by K positions

```
#include<stdio.h>
int main(){
int n,k,i,j,temp;
scanf("%d%d",&n,&k);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
k%=n;
for(j=0;j<k;j++){
```

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```
temp=a[n-1];
for(i=n-1;i>0;i--)a[i]=a[i-1];
a[0]=temp;
}
for(i=0;i<n;i++)printf("%d ",a[i]);
}
```

52. Reverse array in groups of K

```
#include<stdio.h>
int main(){
int n,k,i,j,temp;
scanf("%d%d",&n,&k);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i+=k){
j=i+k-1;
if(j>=n)j=n-1;
while(i<j){
temp=a[i];
a[i]=a[j];
a[j]=temp;
i++;
j--;
}
}
for(i=0;i<n;i++)printf("%d ",a[i]);
}
```

53. Kth largest element

```
#include<stdio.h>
int main(){
int n,k,i,j,temp;
scanf("%d%d",&n,&k);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n-1;i++){
for(j=i+1;j<n;j++){
if(a[i]<a[j]){
temp=a[i];
a[i]=a[j];
a[j]=temp;
}
}
}
printf("%d",a[k-1]);
}
```

54. Kth smallest element

```
#include<stdio.h>
int main(){
int n,k,i,j,temp;
scanf("%d%d",&n,&k);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n-1;i++){
for(j=i+1;j<n;j++){
if(a[i]>a[j]){
temp=a[i];
a[i]=a[j];
a[j]=temp;
}
}
}
printf("%d",a[k-1]);
}
```

55. Top K largest elements

```
#include<stdio.h>
int main(){
int n,k,i,j,temp;
scanf("%d%d",&n,&k);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n-1;i++){
for(j=i+1;j<n;j++){
if(a[i]<a[j]){
temp=a[i];
a[i]=a[j];
a[j]=temp;
}
}
}
for(i=0;i<k;i++)printf("%d ",a[i]);
}
```

56. Top K smallest elements

```
#include<stdio.h>
int main(){
int n,k,i,j,temp;
scanf("%d%d",&n,&k);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n-1;i++){
for(j=i+1;j<n;j++){
if(a[i]>a[j]){
temp=a[i];
a[i]=a[j];
a[j]=temp;
}
}
}
for(i=0;i<k;i++)printf("%d ",a[i]);
}
```

57. Merge two sorted arrays

```
#include<stdio.h>
int main(){
int n,m,i,j,k=0;
```

```
scanf("%d",&n);
int a[n],b[m+n],c[n+m];
for(i=0;i<n;i++)scanf("%d",&a[i]);
scanf("%d",&m);
for(i=0;i<m;i++)scanf("%d",&b[i]);
i=0;
j=0;
while(i<n&&j<m){
if(a[i]<b[j])c[k++]=a[i++];
else c[k++]=b[j++];
}
while(i<n)c[k++]=a[i++];
while(j<m)c[k++]=b[j++];
for(i=0;i<k;i++)printf("%d ",c[i]);
}
```

58. Merge two arrays without using an extra array

```
#include<stdio.h>
int main(){
int n,m,i,j,temp;
scanf("%d%d",&n,&m);
int a[n+m],b[m];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<m;i++)scanf("%d",&b[i]);
for(i=0;i<m;i++)a[n+i]=b[i];
for(i=0;i<n+m-1;i++){
for(j=i+1;j<n+m;j++){
if(a[i]>a[j]){
temp=a[i];
a[i]=a[j];
a[j]=temp;
}
}
}
for(i=0;i<n+m;i++)printf("%d ",a[i]);
}
```

59. Median of an array

```
#include<stdio.h>
int main(){
int n,i,j,temp;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n-1;i++){
for(j=i+1;j<n;j++){
if(a[i]>a[j]){
temp=a[i];
a[i]=a[j];
a[j]=temp;
}
}
}
if(n%2)printf("%.2f",(float)a[n/2]);
else printf("%.2f",(a[n/2-1]+a[n/2])/2.0);
}
```

60. Median of two sorted arrays

```
#include<stdio.h>
int main(){
int n,m,i,j,k=0;
scanf("%d",&n);
int a[n],b[m],c[n+m];
for(i=0;i<n;i++)scanf("%d",&a[i]);
scanf("%d",&m);
for(i=0;i<m;i++)scanf("%d",&b[i]);
i=0;
j=0;
while(i<n&&j<m){
if(a[i]<b[j])c[k++]=a[i++];
else c[k++]=b[j++];
}
while(i<n)c[k++]=a[i++];
while(j<m)c[k++]=b[j++];
if(k%2)printf("%.2f",(float)c[k/2]);
else printf("%.2f",(c[k/2-1]+c[k/2])/2.0);
}
```

LEVEL 4: SUBARRAY PROBLEMS

61. Maximum sum subarray

```
#include<stdio.h>
int main(){
int n,i,sum,max;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
sum=max=a[0];
for(i=1;i<n;i++){
if(sum+a[i]<a[i])sum=a[i];
else sum+=a[i];
if(sum>max)max=sum;
}
printf("%d",max);
}
```

62. Minimum sum subarray

```
#include<stdio.h>
int main(){
int n,i,sum,min;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
sum=min=a[0];
for(i=1;i<n;i++){
if(sum+a[i]>a[i])sum=a[i];
else sum+=a[i];
if(sum<min)min=sum;
```

```
}
printf("%d",min);
}
```

63. Maximum product subarray

```
#include<stdio.h>
int main(){
int n,i,max,min,temp,result;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
max=min=result=a[0];
for(i=1;i<n;i++){
if(a[i]<0){
temp=max;
max=min;
min=temp;
}
if(a[i]>max*a[i])max=a[i];
else max=max*a[i];
if(a[i]<min*a[i])min=a[i];
else min=min*a[i];
if(max>result)result=max;
}
printf("%d",result);
}
```

64. Minimum product subarray

```
#include<stdio.h>
int main(){
int n,i,max,min,temp,result;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
max=min=result=a[0];
for(i=1;i<n;i++){
if(a[i]<0){
temp=max;
max=min;
min=temp;
}
if(a[i]>max*a[i])max=a[i];
else max=max*a[i];
if(a[i]<min*a[i])min=a[i];
else min=min*a[i];
if(min<result)result=min;
}
printf("%d",result);
}
```

65. Longest subarray with given sum

```
#include<stdio.h>
int main(){
int n,i,j,sum,target,length=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
scanf("%d",&target);
for(i=0;i<n;i++){
sum=0;
for(j=i;j<n;j++){
sum+=a[j];
if(sum==target&&j-i+1>length)length=j-i+1;
}
}
printf("%d",length);
}
```

66. Shortest subarray with given sum

```
#include<stdio.h>
int main(){
int n,i,j,sum,target,length=n+1;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
scanf("%d",&target);
for(i=0;i<n;i++){
sum=0;
for(j=i;j<n;j++){
sum+=a[j];
if(sum==target&&j-i+1<length)length=j-i+1;
}
}
if(length==n+1)length=0;
printf("%d",length);
}
```

67. Count subarrays having given sum

```
#include<stdio.h>
int main(){
int n,i,j,sum,target,count=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
scanf("%d",&target);
for(i=0;i<n;i++){
sum=0;
for(j=i;j<n;j++){
sum+=a[j];
if(sum==target)count++;
}
}
printf("%d",count);
}
```

68. Find a subarray whose sum is zero

```
#include<stdio.h>
int main(){
```

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```
int n,i,j,sum;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
sum=0;
for(j=i;j<n;j++){
sum+=a[j];
if(sum==0){
for(;i<=j;i++)printf("%d ",a[i]);
return 0;
}
}
}
printf("No subarray");
}
```

69. Count subarrays whose sum is zero

```
#include<stdio.h>
int main(){
int n,i,j,sum,count=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
sum=0;
for(j=i;j<n;j++){
sum+=a[j];
if(sum==0)count++;
}
}
printf("%d",count);
}
```

70. Longest subarray with sum zero

```
#include<stdio.h>
int main(){
int n,i,j,sum,length=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
sum=0;
for(j=i;j<n;j++){
sum+=a[j];
if(sum==0&&j-i+1>length)length=j-i+1;
}
}
printf("%d",length);
}
```

71. Maximum sum subarray of size K

```
#include<stdio.h>
int main(){
int n,k,i,sum=0,max;
scanf("%d%d",&n,&k);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<k;i++)sum+=a[i];
max=sum;
for(i=k;i<n;i++){
sum=sum+a[i]-a[i-k];
if(sum>max)max=sum;
}
printf("%d",max);
}
```

72. Minimum sum subarray of size K

```
#include<stdio.h>
int main(){
int n,k,i,sum=0,min;
scanf("%d%d",&n,&k);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<k;i++)sum+=a[i];
min=sum;
for(i=k;i<n;i++){
sum=sum+a[i]-a[i-k];
if(sum<min)min=sum;
}
printf("%d",min);
}
```

73. First negative number in every window of size K

```
#include<stdio.h>
int main(){
int n,k,i,j,found;
scanf("%d%d",&n,&k);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<=n-k;i++){
found=0;
for(j=i;j<i+k;j++){
if(a[j]<0){
printf("%d ",a[j]);
found=1;
break;
}
}
if(!found)printf("0 ");
}
}
```

74. Maximum element in every window of size K

```
#include<stdio.h>
int main(){
int n,k,i,j,max;
scanf("%d%d",&n,&k);
```

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```
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<=n-k;i++){
max=a[i];
for(j=i+1;j<i+k;j++){
if(a[j]>max)max=a[j];
}
printf("%d ",max);
}
}
```

75. Minimum element in every window of size K

```
#include<stdio.h>
int main(){
int n,k,i,j,min;
scanf("%d%d",&n,&k);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<=n-k;i++){
min=a[i];
for(j=i+1;j<i+k;j++){
if(a[j]<min)min=a[j];
}
printf("%d ",min);
}
}
```

76. Longest subarray with equal 0s and 1s

```
#include<stdio.h>
int main(){
int n,i,j,zero,one,length=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
zero=one=0;
for(j=i;j<n;j++){
if(a[j]==0)zero++;
else one++;
if(zero==one&&j-i+1>length)length=j-i+1;
}
}
printf("%d",length);
}
```

77. Longest subarray with equal positive and negative elements

```
#include<stdio.h>
int main(){
int n,i,j,p,ne,length=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
p=ne=0;
for(j=i;j<n;j++){
if(a[j]>0)p++;
else if(a[j]<0)ne++;
if(p==ne&&j-i+1>length)length=j-i+1;
}
}
printf("%d",length);
}
```

78. Longest subarray without duplicate elements

```
#include<stdio.h>
int main(){
int n,i,j,k,found,length=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
for(j=i;j<n;j++){
found=0;
for(k=i;k<j;k++){
if(a[k]==a[j]){
found=1;
break;
}
}
if(found)break;
if(j-i+1>length)length=j-i+1;
}
}
printf("%d",length);
}
```

79. Longest consecutive sequence

```
#include<stdio.h>
int main(){
int n,i,j,count,best=0,found;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
count=1;
j=a[i]+1;
while(1){
found=0;
for(int k=0;k<n;k++){
if(a[k]==j){
found=1;
break;
}
}
if(!found)break;
count++;
```

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```
j++;
}
if(count>best)best=count;
}
printf("%d",best);
}
```

80. Maximum length of consecutive 1s

```
#include<stdio.h>
int main(){
int n,i,count=0,max=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
if(a[i]==1){
count++;
if(count>max)max=count;
}else count=0;
}
printf("%d",max);
}
```

LEVEL 5: ADVANCED ARRAY PROBLEMS

81. Equilibrium index

```
#include<stdio.h>
int main(){
int n,i,total=0,left=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++){
scanf("%d",&a[i]);
total+=a[i];
}
for(i=0;i<n;i++){
total-=a[i];
if(left==total){
printf("%d",i);
return 0;
}
left+=a[i];
}
printf("-1");
}
```

82. All equilibrium indices

```
#include<stdio.h>
int main(){
int n,i,total=0,left=0,found=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++){
scanf("%d",&a[i]);
total+=a[i];
}
for(i=0;i<n;i++){
total-=a[i];
if(left==total){
printf("%d ",i);
found=1;
}
left+=a[i];
}
if(!found)printf("-1");
}
```

83. Product of array elements except itself

```
#include<stdio.h>
int main(){
int n,i,j,product;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
product=1;
for(j=0;j<n;j++){
if(i!=j)product*=a[j];
}
printf("%d ",product);
}
}
```

84. Maximum profit from buying and selling once

```
#include<stdio.h>
int main(){
int n,i,min,profit=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
min=a[0];
for(i=1;i<n;i++){
if(a[i]-min>profit)profit=a[i]-min;
if(a[i]<min)min=a[i];
}
printf("%d",profit);
}
```

85. Maximum profit with multiple transactions

```
#include<stdio.h>
int main(){
int n,i,profit=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=1;i<n;i++){
if(a[i]>a[i-1])profit+=a[i]-a[i-1];
}
printf("%d",profit);
```

```
}
```

86. Trapped rainwater

```
#include<stdio.h>
int main(){
int n,i,j,left,right,water=0,min;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
left=right=0;
for(j=0;j<=i;j++)if(a[j]>left)left=a[j];
for(j=i;j<n;j++)if(a[j]>right)right=a[j];
min=left<right?left:right;
water+=min-a[i];
}
printf("%d",water);
}
```

87. Maximum area of water

```
#include<stdio.h>
int main(){
int n,i,j,area,max=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
for(j=i+1;j<n;j++){
area=(j-i)*(a[i]<a[j]?a[i]:a[j]);
if(area>max)max=area;
}
}
printf("%d",max);
}
```

88. Next greater element

```
#include<stdio.h>
int main(){
int n,i,j,found;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
found=0;
for(j=i+1;j<n;j++){
if(a[j]>a[i]){
printf("%d ",a[j]);
found=1;
break;
}
}
if(!found)printf("-1 ");
}
}
```

89. Next smaller element

```
#include<stdio.h>
int main(){
int n,i,j,found;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
found=0;
for(j=i+1;j<n;j++){
if(a[j]<a[i]){
printf("%d ",a[j]);
found=1;
break;
}
}
if(!found)printf("-1 ");
}
}
```

90. Previous greater element

```
#include<stdio.h>
int main(){
int n,i,j,found;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
found=0;
for(j=i-1;j>=0;j--){
if(a[j]>a[i]){
printf("%d ",a[j]);
found=1;
break;
}
}
if(!found)printf("-1 ");
}
}
```

91. Previous smaller element

```
#include<stdio.h>
int main(){
int n,i,j,found;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
found=0;
for(j=i-1;j>=0;j--){
if(a[j]<a[i]){
printf("%d ",a[j]);
found=1;
```

```
break;
}
}
if(!found)printf("-1 ");
}
}
```

92. Leaders in an array

```
#include<stdio.h>
int main(){
int n,i,j,leader;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
leader=1;
for(j=i+1;j<n;j++){
if(a[i]<a[j]){
leader=0;
break;
}
}
if(leader)printf("%d ",a[i]);
}
}
```

93. Maximum circular subarray sum

```
#include<stdio.h>
int main(){
int n,i,sum=0,maxsum,minsum,maxend,minend;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
maxend=maxsum=a[0];
minend=minsum=a[0];
for(i=1;i<n;i++){
if(maxend+a[i]<a[i])maxend=a[i];
else maxend+=a[i];
if(maxend>maxsum)maxsum=maxend;
if(minend+a[i]>a[i])minend=a[i];
else minend+=a[i];
if(minend<minsum)minsum=minend;
sum+=a[i];
}
sum+=a[0];
if(maxsum<0)printf("%d",maxsum);
else printf("%d",sum-minsum);
}
```

94. Minimum jumps to reach last index

```
#include<stdio.h>
int main(){
int n,i,j,jumps=0,pos=0,maxreach;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
while(pos<n-1){
maxreach=pos;
for(j=pos+1;j<=pos+a[pos]&&j<n;j++){
if(j+a[j]>maxreach+a[maxreach])maxreach=j;
}
if(maxreach==pos){
printf("-1");
return 0;
}
pos=maxreach;
jumps++;
}
printf("%d",jumps);
}
```

95. Check whether last index is reachable

```
#include<stdio.h>
int main(){
int n,i,maxreach=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
if(i>maxreach)break;
if(i+a[i]>maxreach)maxreach=i+a[i];
}
if(maxreach>=n-1)printf("Yes");
else printf("No");
}
```

96. Longest increasing subsequence

```
#include<stdio.h>
int main(){
int n,i,j,max=1;
scanf("%d",&n);
int a[n],dp[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++)dp[i]=1;
for(i=1;i<n;i++){
for(j=0;j<i;j++){
if(a[i]>a[j]&&dp[i]<dp[j]+1)dp[i]=dp[j]+1;
}
if(dp[i]>max)max=dp[i];
}
printf("%d",max);
}
```

97. Longest decreasing subsequence

```
#include<stdio.h>
int main(){
int n,i,j,max=1;
scanf("%d",&n);
```

```
int a[n],dp[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++)dp[i]=1;
for(i=1;i<n;i++){
for(j=0;j<i;j++){
if(a[i]<a[j]&&dp[i]<dp[j]+1)dp[i]=dp[j]+1;
}
if(dp[i]>max)max=dp[i];
}
printf("%d",max);
}
```

98. Maximum sum increasing subsequence

```
#include<stdio.h>
int main(){
int n,i,j,max;
scanf("%d",&n);
int a[n],sum[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++)sum[i]=a[i];
for(i=1;i<n;i++){
for(j=0;j<i;j++){
if(a[i]>a[j]&&sum[i]<sum[j]+a[i])sum[i]=sum[j]+a[i];
}
}
max=sum[0];
for(i=1;i<n;i++)if(sum[i]>max)max=sum[i];
printf("%d",max);
}
```

99. Number of inversions

```
#include<stdio.h>
int main(){
int n,i,j,count=0;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++){
for(j=i+1;j<n;j++){
if(a[i]>a[j])count++;
}
}
printf("%d",count);
}
```

100. Maximum subarray sum after deleting at most one element

```
#include<stdio.h>
int main(){
int n,i,maxend,delend,maxsum;
scanf("%d",&n);
int a[n];
for(i=0;i<n;i++)scanf("%d",&a[i]);
maxend=a[0];
delend=-2147483647;
maxsum=a[0];
for(i=1;i<n;i++){
if(delend+a[i]<maxend)delend=maxend;
else delend=delend+a[i];
if(maxend+a[i]<a[i])maxend=a[i];
else maxend=maxend+a[i];
if(delend>maxsum)maxsum=delend;
if(maxend>maxsum)maxsum=maxend;
}
printf("%d",maxsum);
}
```

Suraj